

Supplementary File S2

RQ2: Reinforcement-Learning Formulation Consistency and Cross-Study Comparability

Supporting material for *What the Survey Literature Says About Reinforcement Learning for Building Control: A Critical Synthesis*

Scope and coding scale

This supplementary file provides the survey-level evidence audit for RQ2. It reports formulation-consistency scores for state, action, reward, objectives, and constraints; identifies comparability barriers; and presents survey-level evidence justifications, summaries, and concise answers.

Score	Evidence-intensity interpretation
0	Not addressed
1	Mentioned
2	Discussed
3	Structuring role

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S2.1 Reinforcement learning for building controls: The opportunities and challenges [1]

Table S2.1: RQ2 recoded audit table for Wang & Hong (2020). Scores use the shared 0–3 evidence-intensity scale.

Survey	State	Action	Reward	Objectives	Constraints	Comparability problems	Recoded evidence justification
Wang and Hong (2020)	3	3	3	2	2	3	The survey visibly structures its review around core RL components, especially algorithms, states, actions, rewards, and environment. State, action, and reward therefore receive score 3 because they organize dedicated parts of the paper. Objectives receive score 2 because building-control goals such as energy, comfort, demand flexibility, robustness, and generalization are discussed, but they do not form a standalone objective taxonomy. Constraints receive score 2 because safety, robustness, control security, comfort, and deployment limitations are discussed, but constraints are not treated as a separate systematic coding dimension. Comparability problems receive score 3 because the paper explicitly identifies inconsistent baselines and environmental settings as major obstacles to cross-study comparison and calls for open, shared, and well-recognized control testbeds.

Table S2.2: RQ2 formulation-consistency scores for Wang & Hong (2020), recoded using the shared evidence-intensity scale.

Survey	State	Action	Reward	Objectives	Constraints
Wang & Hong (2020)	3	3	3	2	2

Table S2.3: RQ2 comparability-barrier scores for Wang & Hong (2020), recoded using the shared evidence-intensity scale.

Survey	State	Action	Reward	Objective/metric	Constraint	Baseline	Simulator/testbed
Wang & Hong (2020)	2	2	2	2	2	3	3

Table S2.4: RQ2 summary table for Wang & Hong (2020), after recoding.

Survey	Formulation strength	Biggest inconsistency identified	Main comparability consequence
Wang & Hong (2020)	Strong	Inconsistent specification of states, actions, rewards, environments, and evaluation baselines across studies	Cross-study comparison becomes difficult because studies use different RL formulations, baseline controllers, and environmental settings; this motivates shared testbeds, open datasets, and more standardized benchmarking.

Table S2.5: One-line RQ2 answer for Wang & Hong (2020), after recoding.

Survey	RQ2 answer
Wang and Hong (2020)	Wang & Hong (2020) answer RQ2 strongly by making RL formulation itself a structuring principle of the review, especially through state, action, reward, and environment. The paper also shows that cross-study comparison is weakened by inconsistent formulation choices, uneven baselines, and different environmental settings, leading to a clear call for open and shared control testbeds.

S2.2 The reinforcement learning method for occupant behavior in building control: A review [2]

Table S2.6: RQ2 recoded audit table for Han et al. (2021). Scores use the shared 0–3 evidence-intensity scale.

Survey	State	Action	Reward	Objectives	Constraints	Comparability problems	Recoded evidence justification
Han et al. (2021)	3	2	3	2	0	2	The survey gives visible structural attention to how occupant behavior enters the MDP. State receives score 3 because occupant behavior as state or state-transition influence is a central organizing route in the review. Reward receives score 3 because the paper explicitly identifies reward as highly sensitive to occupant behavior and comfort preferences, and treats reward as a main route through which occupant behavior affects the MDP. Action receives score 2 because action and occupant feedback are discussed, but only a smaller subset of reviewed studies connects occupant behavior to action. Objectives receive score 2 because comfort, energy, indoor air quality, lighting, and occupant-centered goals are discussed, but the paper does not organize itself around an objective taxonomy. Constraints receive score 0 because constraints are not treated as a visible formulation dimension. Comparability problems receive score 2 because the paper explains that occupant behavior can be represented through different MDP routes, which weakens formulation consistency, but it does not develop this into a broader benchmarking or reproducibility agenda.

Table S2.7: RQ2 formulation-consistency scores for Han et al. (2021), recoded using the shared evidence-intensity scale.

Survey	State	Action	Reward	Objectives	Constraints
Han et al. (2021)	3	2	3	2	0

Table S2.8: RQ2 comparability-barrier scores for Han et al. (2021), recoded using the shared evidence-intensity scale.

Survey	State	Action	Reward	Objective/metric	Constraint	Baseline	Simulator/testbed
Han et al. (2021)	2	1	2	2	0	0	0

Table S2.9: Occupancy- and occupant-focused formulation routes identified by Han et al. (2021).

Route	MDP component affected	Interpretation for RQ2
Occupant behavior as state	State	Occupant behavior is frequently represented as part of the state, especially in HVAC control. This makes state design dependent on how occupancy and behavior are measured and encoded.
Occupancy or actual behavior affecting dynamics	State transition / environment	Some studies represent occupant behavior indirectly through occupancy or actual behavior that changes the building state transition. This means that similar occupant-aware control problems may place the same phenomenon in different parts of the MDP.
Occupant behavior affecting reward	Reward	Reward design is especially sensitive to occupant behavior because comfort preferences and behavioral profiles vary across occupants and contexts. This makes reward design difficult to standardize across studies.
Occupant feedback	Reward / occasional action-related interaction	Some studies use feedback from occupants to shape learning or update control behavior. This introduces another formulation route that differs from occupancy-as-state formulations.

Table S2.10: RQ2 summary table for Han et al. (2021), after recoding.

Survey	Formulation strength	Biggest inconsistency identified	Main comparability consequence
Han et al. (2021)	Moderate but focused	Occupant behavior is formulated inconsistently across studies, appearing as state information, a state-transition influence, a reward modifier, or a feedback signal.	Studies that appear to address similar occupant-aware building-control problems may not be solving the same RL formulation, which weakens cross-study comparability.

Table S2.11: One-line RQ2 answer for Han et al. (2021), after recoding.

Survey	RQ2 answer
Han et al. (2021)	Han et al. (2021) answer RQ2 by showing that RL formulation becomes fragmented when occupant behavior is included. Occupant-related information can enter the MDP as state information, as an influence on state transition, as a reward modifier, or as feedback to the control system. This makes occupant-aware RL studies difficult to compare because apparently similar applications may rely on different underlying MDP formulations.

S2.3 Systematic review on deep reinforcement learning-based energy management for different building types [3]

Table S2.12: RQ2 recoded audit table for Shaqour & Hagishima (2022). Scores use the shared 0–3 evidence-intensity scale.

Survey	State	Action	Reward	Objectives	Constraints	Comparability problems	Recoded evidence justification
Shaqour and Hagishima (2022)	2	2	2	3	2	3	The survey provides a paragraph-level MDP and DRL background in which states, actions, rewards, transition probability, and policy are explained, so state, action, and reward receive score 2 rather than 1. These components are discussed conceptually, but they do not structure the cross-study review as standalone formulation taxonomies. Objectives receive score 3 because the paper’s comparative analysis is visibly organized around building types, controlled systems, renewable-energy integration, demand response, and objectives beyond energy, such as cost and comfort. Constraints receive score 2 because the paper discusses multidimensional operational objectives and constraints in BEMS, including comfort, energy/cost, data availability, and implementation limits, but does not develop constraints as a separate systematic taxonomy. Comparability problems receive score 3 because the paper explicitly emphasizes data scarcity, limited real-system validation, lack of open-source code, lack of standardized datasets, and the need for more reproducible and comparable DRL-based BEMS research.

Table S2.13: RQ2 formulation-consistency scores for Shaqour & Hagishima (2022), recoded using the shared evidence-intensity scale.

Survey	State	Action	Reward	Objectives	Constraints
Shaqour & Hagishima (2022)	2	2	2	3	2

Table S2.14: RQ2 comparability-barrier scores for Shaqour & Hagishima (2022), recoded using the shared evidence-intensity scale.

Survey	State	Action	Reward	Objective/metric	Constraint	Baseline	Simulator/testbed
Shaqour & Hagishima (2022)	0	0	1	3	2	2	3

Table S2.15: RQ2 summary table for Shaqour & Hagishima (2022), after recoding.

Survey	Formulation strength	Biggest inconsistency identified	Main comparability consequence
Shaqour & Hagishima (2022)	Moderate, objective-centered	DRL-based BEMS studies differ across building types in controlled systems, objectives, constraints, data availability, validation setting, and system configuration.	Cross-study comparison is weakened because reported performance depends on building type, objective mix, available data, baseline/controller setting, and whether the work is simulation-only or validated in a real system. This motivates open data, open-source code, and more standardized validation environments.

Table S2.16: One-line RQ2 answer for Shaqour & Hagishima (2022), after recoding.

Survey	RQ2 answer
Shaqour and Hagishima (2022)	Shaqour & Hagishima (2022) contribute to RQ2 mainly by showing that DRL-based BEMS studies vary strongly by building type, objective mix, controlled systems, data availability, and validation setting. The paper gives a basic MDP formulation of state, action, and reward, but its strongest contribution is to comparability and reproducibility: it highlights limited real-world validation, lack of open-source code, lack of standardized datasets, and the need for more comparable environments such as CityLearn.

S2.4 Reinforcement learning for optimizing renewable energy utilization in buildings: A review on applications and innovations [4]

Table S2.17: RQ2 recoded audit table for Michailidis et al. (2025). Scores use the shared 0–3 evidence-intensity scale.

Survey	State	Action	Reward	Objectives	Constraints	Comparability problems	Recoded evidence justification
Michailidis et al. (2025)	2	2	3	3	2	3	The survey gives a generic RL/BEMS formulation and control-loop discussion, so state and action receive score 2: they are explained as part of RL control, but they are not the main cross-study taxonomy. Reward receives score 3 because reward functions are one of the explicit evaluation attributes and are synthesized through a dedicated reward-function discussion and occurrence figure. Objectives receive score 3 because the paper structurally organizes target outputs and evaluation metrics, including comfort, energy savings, cost reduction, renewable-energy use, grid reliance, environmental footprint, computational efficiency, scalability, and data efficiency. Constraints receive score 2 because real-time constraints, grid reliance, RES variability, comfort–cost trade-offs, and implementation limitations are discussed, but constraints are not treated as a standalone systematic formulation taxonomy. Comparability problems receive score 3 because the paper structurally addresses uneven baselines, heterogeneous metrics, and simulation-heavy validation through baseline-control discussion, target-output synthesis, and simulation/real-life summary tables.

Table S2.18: RQ2 formulation-consistency scores for Michailidis et al. (2025), recoded using the shared evidence-intensity scale.

Survey	State	Action	Reward	Objectives	Constraints
Michailidis et al. (2025)	2	2	3	3	2

Table S2.19: RQ2 comparability-barrier scores for Michailidis et al. (2025), recoded using the shared evidence-intensity scale.

Survey	State	Action	Reward	Objective/metric	Constraint	Baseline	Simulator/testbed
Michailidis et al. (2025)	0	0	2	3	2	3	3

Table S2.20: RQ2 summary table for Michailidis et al. (2025), after recoding.

Survey	Formulation strength	Biggest inconsistency identified	Main comparability consequence
Michailidis et al. (2025)	Strong for rewards, objectives, baselines, and validation setting	RES-integrated BEMS studies differ in reward-function design, target outputs, baseline-control practice, metric selection, and simulation versus real-life validation status.	Cross-study comparison is weakened because studies optimize different mixes of comfort, cost, energy savings, renewable utilization, grid reliance, and environmental metrics, while relying on uneven baselines and mostly simulation-based validation.

Table S2.21: One-line RQ2 answer for Michailidis et al. (2025), after recoding.

Survey	RQ2 answer
Michailidis et al. (2025)	Michailidis et al. (2025) contribute to RQ2 mainly by structurally organizing reward-function types, target outputs, baseline-control practices, and simulation versus real-life validation across RL studies for RES-integrated BEMS. The paper only partially treats state and action formulation, but it strongly shows that heterogeneous metrics, uneven baselines, multi-objective trade-offs, and simulation-heavy validation hinder meaningful cross-study comparison.

S2.5 Reinforcement learning for HVAC control in intelligent buildings: A technical and conceptual review [5]

Table S2.22: RQ2 recoded audit table for Al Sayed et al. (2024). Scores use the shared 0–3 evidence-intensity scale.

Survey	State	Action	Reward	Objectives	Constraints	Comparability problems	Recoded evidence justification
Al Sayed et al. (2024)	3	3	3	3	2	3	The survey has a dedicated RL modeling-environment section and visibly structures its review around state, action, reward, probability/model environment, and algorithm implementation. State receives score 3 because the paper distinguishes endogenous and exogenous state components and summarizes them across studies. Action receives score 3 because HVAC actions are organized across source, condenser, AHU, and terminal components. Reward receives score 3 because reward formulation is treated in a dedicated subsection and summarized across comfort, energy, cost, air-quality, and other components. Objectives receive score 3 because energy reduction, cost minimization, thermal comfort, air quality, and related control targets are structurally embedded in the reward/objective synthesis. Constraints receive score 2 because comfort thresholds, CO ₂ limits, penalties, safety, and deployment restrictions are discussed, but constraints are not developed as a standalone taxonomy. Comparability problems receive score 3 because the paper strongly emphasizes the mismatch between virtual training environments and real deployment, especially static predefined exogenous-state sequences, limited real-building implementation, and simulator-to-reality transfer limitations.

Table S2.23: RQ2 formulation-consistency scores for Al Sayed et al. (2024), recoded using the shared evidence-intensity scale.

Survey	State	Action	Reward	Objectives	Constraints
Al Sayed et al. (2024)	3	3	3	3	2

Table S2.24: RQ2 comparability-barrier scores for Al Sayed et al. (2024), recoded using the shared evidence-intensity scale.

Survey	State	Action	Reward	Objective/metric	Constraint	Baseline	Simulator/testbed
Al Sayed et al. (2024)	3	0	2	2	2	0	3

Table S2.25: RQ2 summary table for Al Sayed et al. (2024), after recoding.

Survey	Formulation strength	Biggest inconsistency identified	Main comparability consequence
Al Sayed et al. (2024)	Strong	Mismatch between how state dynamics, especially exogenous variables, are represented in virtual training environments and how they evolve in real buildings	Policies that perform well in simulation may transfer poorly or unsafely to deployment, making cross-study comparison less meaningful when studies rely on different simulators, different exogenous-state assumptions, and limited real-world validation.

Table S2.26: One-line RQ2 answer for Al Sayed et al. (2024), after recoding.

Survey	RQ2 answer
Al Sayed et al. (2024)	Al Sayed et al. (2024) contribute strongly to RQ2 by structurally organizing HVAC RL studies around state, action, reward, and objective formulation. The paper also shows that comparability and safe transfer are weakened by the gap between virtual training environments and real operating conditions, especially exogenous-state variation, static training sequences, simulator dependence, and limited real-building deployment.

S2.6 A review of deep reinforcement learning for smart building energy management [6]

Table S2.27: RQ2 recoded audit table for Yu et al. (2021). Scores use the shared 0–3 evidence-intensity scale.

Survey	State	Action	Reward	Objectives	Constraints	Comparability problems	Recoded evidence justification
Yu et al. (2021)	2	2	2	3	2	3	The survey provides generic DRL/MDP and SBEM formulation, so state, action, and reward receive score 2: they are explained and illustrated, but they do not structure the review as standalone formulation taxonomies. Objectives receive score 3 because the paper systematically synthesizes optimization objectives across single building subsystems, multiple building subsystems, and building microgrids, especially through its summary tables and system-scale organization. Constraints receive score 2 because uncertain parameters, spatially and temporally coupled operational constraints, indoor temperature constraints, CO ₂ constraints, and physical system limits are discussed, but constraints are not treated as an independent taxonomy. Comparability problems receive score 3 because the paper repeatedly emphasizes that most DRL-based SBEM evidence remains simulation-based, that practical deployment is limited, and that model-free DRL methods are rarely implemented in real systems despite reported performance gains.

Table S2.28: RQ2 formulation-consistency scores for Yu et al. (2021), recoded using the shared evidence-intensity scale.

Survey	State	Action	Reward	Objectives	Constraints
Yu et al. (2021)	2	2	2	3	2

Table S2.29: RQ2 comparability-barrier scores for Yu et al. (2021), recoded using the shared evidence-intensity scale.

Survey	State	Action	Reward	Objective/metric	Constraint	Baseline	Simulator/testbed
Yu et al. (2021)	0	1	2	3	2	0	3

Table S2.30: RQ2 summary table for Yu et al. (2021), after recoding.

Survey	Formulation strength	Biggest inconsistency identified	Main comparability consequence
Yu et al. (2021)	Moderate, objective-centered	SBEM studies optimize different combinations of cost, energy consumption, comfort, air quality, carbon emissions, peak demand, and other goals, usually without a unified formulation framework.	Cross-study comparison is weakened because studies differ in objective mix, reward weighting, constraint handling, system scale, and implementation setting, while most reported results still come from simulations rather than practical deployment.

Table S2.31: One-line RQ2 answer for Yu et al. (2021), after recoding.

Survey	RQ2 answer
Yu et al. (2021)	Yu et al. (2021) contribute to RQ2 mainly by structurally synthesizing objectives across different smart-building energy-management scales and by exposing implementation and comparability gaps. The paper discusses state, action, reward, and constraints through generic DRL/SBEM formulation, but its strongest evidence concerns objective heterogeneity, multi-objective reward weighting, simulation-heavy validation, and limited real-world deployment.

S2.7 A review of reinforcement learning applications to control of heating, ventilation and air conditioning systems [7]

Table S2.32: RQ2 recoded audit table for Sierla et al. (2022). Scores use the shared 0–3 evidence-intensity scale.

Survey	State	Action	Reward	Objectives	Constraints	Comparability problems	Recoded evidence justification
Sierla et al. (2022)	2	3	2	3	2	3	The survey contributes to RQ2 most strongly through action-space formulation and indoor-environment target variables. State receives score 2 because the paper discusses state as part of the RL environment and HVAC-control formulation, but does not systematically classify state choices across studies. Action receives score 3 because the paper explicitly aims to identify action formulations in the literature and categorizes studies by binary, discrete, and continuous action types. Reward receives score 2 because reward is discussed as feedback that captures HVAC-control objectives, but reward is not the main taxonomy. Objectives receive score 3 because the review categorizes studies by controlled indoor-environment aspects, especially temperature, humidity, and air quality. Constraints receive score 2 because comfort intervals, indoor-environment requirements, backup-controller issues, deployability, and action-space scalability are discussed, but constraints are not developed as a standalone taxonomy. Comparability problems receive score 3 because the paper makes action abstraction, mapping to real HVAC equipment, virtual-environment design, and deployment readiness central to its critique of the literature.

Table S2.33: RQ2 formulation-consistency scores for Sierla et al. (2022), recoded using the shared evidence-intensity scale.

Survey	State	Action	Reward	Objectives	Constraints
Sierla et al. (2022)	2	3	2	3	2

Table S2.34: RQ2 comparability-barrier scores for Sierla et al. (2022), recoded using the shared evidence-intensity scale.

Survey	State	Action	Reward	Objective/metric	Constraint	Baseline	Simulator/testbed
Sierla et al. (2022)	2	3	2	3	2	0	3

Table S2.35: RQ2 summary table for Sierla et al. (2022), after recoding.

Survey	Formulation strength	Biggest inconsistency identified	Main comparability consequence
Sierla et al. (2022)	Strong but specialized	HVAC RL studies vary widely in action-space design, control abstraction, and indoor-environment targets, especially temperature-only versus multi-variable formulations including humidity and air quality.	Cross-study comparison and deployment assessment become difficult because studies often use different action abstractions, optimize different target variables, and provide insufficiently clear mappings between RL decisions, virtual environments, and real HVAC system interfaces.

Table S2.36: One-line RQ2 answer for Sierla et al. (2022), after recoding.

Survey	RQ2 answer
Sierla et al. (2022)	Sierla et al. (2022) contribute to RQ2 mainly by structurally showing that inconsistency in HVAC RL studies is concentrated in action-space formulation, abstraction level, and indoor-environment target variables. These differences make cross-study comparison and real-system deployment harder because RL actions are not always clearly mapped to practical HVAC equipment, and studies often rely on different virtual-environment assumptions.

S2.8 A review of reinforcement learning for controlling building energy systems from a computer science perspective [8]

Table S2.37: RQ2 recoded audit table for Weinberg et al. (2023). Scores use the shared 0–3 evidence-intensity scale.

Survey	State	Action	Reward	Objectives	Constraints	Comparability problems	Recoded evidence justification
Weinberg et al. (2023)	2	2	2	2	2	3	The survey provides a generic RL formulation for building energy systems, so state, action, and reward receive score 2: they are explained and used in the discussion of RL, model-based RL, sample efficiency, and transfer, but they are not systematically classified as standalone formulation taxonomies. Objectives receive score 2 because control performance, energy use, scalability, sample efficiency, transferability, and multi-agent coordination are discussed, but objectives do not structure the review as an objective taxonomy. Constraints receive score 2 in the formulation matrix because practical limitations such as low-frequency data, long training times, unsafe exploration, communication and interoperability limits, privacy, and scalability are discussed. Comparability problems receive score 3 because the paper structurally organizes computer-science-related challenges for BES control and strongly emphasizes transfer barriers, changing state–action spaces across buildings, sim-to-real mismatch, data limitations, communication constraints, and multi-agent non-stationarity.

Table S2.38: RQ2 formulation-consistency scores for Weinberg et al. (2023), recoded using the shared evidence-intensity scale.

Survey	State	Action	Reward	Objectives	Constraints
Weinberg et al. (2023)	2	2	2	2	2

Table S2.39: RQ2 comparability-barrier scores for Weinberg et al. (2023), recoded using the shared evidence-intensity scale.

Survey	State	Action	Reward	Objective/metric	Constraint	Baseline	Simulator/testbed
Weinberg et al. (2023)	3	3	2	2	3	0	3

Table S2.40: RQ2 summary table for Weinberg et al. (2023), after recoding.

Survey	Formulation strength	Biggest inconsistency identified	Main comparability consequence
Weinberg et al. (2023)	Moderate formulation, strong transfer-barrier analysis	RL-for-BES studies share a generic MDP-level formulation, but practical implementations vary in state-action spaces, training data regimes, simulators, transfer settings, communication assumptions, and multi-agent coordination assumptions.	Cross-study comparison is weakened because policies are trained and evaluated under different building contexts, state-action spaces, simulation assumptions, data frequencies, and coordination structures, making transferability and practical performance hard to assess consistently.

Table S2.41: One-line RQ2 answer for Weinberg et al. (2023), after recoding.

Survey	RQ2 answer
Weinberg et al. (2023)	Weinberg et al. (2023) contribute to RQ2 mainly by showing that RL-for-BES studies share a generic state-action-reward formulation, but differ strongly in the practical conditions that determine transferability and comparability. The paper is especially useful for identifying state-action-space mismatch, sim-to-real mismatch, data limitations, communication and privacy constraints, scalability issues, and multi-agent non-stationarity as barriers to comparing and transferring RL controllers across building contexts.

S2.9 Applications of Deep Reinforcement Learning for Home Energy Management Systems: A Review [9]

Table S2.42: RQ2 recoded audit table for Latoń et al. (2024). Scores use the shared 0–3 evidence-intensity scale.

Survey	State	Action	Reward	Objectives	Constraints	Comparability problems	Recoded evidence justification
Latoń et al. (2024)	2	2	2	3	2	3	The survey provides a substantive RL/MDP background for HEMS, so state, action, and reward receive score 2: they are explained as part of the RL formulation, but they do not structure the review as standalone taxonomies. Objectives receive score 3 because the paper structurally organizes HEMS applications around demand-side response, load scheduling, renewable/storage integration, electric vehicles, cost reduction, comfort, grid stability, and energy efficiency. Constraints receive score 2 because scalability, interoperability, data quality, security, privacy, infrastructure integration, and safety/reliability concerns are discussed, but constraints are not developed as a standalone formulation taxonomy. Comparability problems receive score 3 because the paper strongly emphasizes simulation-heavy validation, limited real-world case studies, benchmarking/KPI needs, digital-twin validation, scalability, interoperability, data/privacy constraints, and the gap between virtual and practical implementation.

Table S2.43: RQ2 formulation-consistency scores for Latoń et al. (2024), recoded using the shared evidence-intensity scale.

Survey	State	Action	Reward	Objectives	Constraints
Latoń et al. (2024)	2	2	2	3	2

Table S2.44: RQ2 comparability-barrier scores for Latoń et al. (2024), recoded using the shared evidence-intensity scale.

Survey	State	Action	Reward	Objective/metric	Constraint	Baseline	Simulator/testbed
Latoń et al. (2024)	0	0	2	3	3	0	3

Table S2.45: RQ2 summary table for Latoń et al. (2024), after recoding.

Survey	Formulation strength	Biggest inconsistency identified	Main comparability consequence
Latoń et al. (2024)	Moderate, objective-centered	HEMS studies optimize different mixes of cost, comfort, energy efficiency, renewable/storage use, demand response, electric-vehicle integration, grid support, and user-oriented goals under heterogeneous validation settings.	Cross-study comparison is weakened because studies differ in objectives, practical assumptions, validation modes, data quality, interoperability, scalability, and privacy/security assumptions, while most evidence still comes from simulations rather than real-world deployments.

Table S2.46: One-line RQ2 answer for Latoń et al. (2024), after recoding.

Survey	RQ2 answer
Latoń et al. (2024)	Latoń et al. (2024) contribute to RQ2 mainly by structurally synthesizing objective heterogeneity across HEMS applications and by highlighting comparability barriers related to simulation-heavy validation, limited real-world case studies, benchmarking needs, scalability, interoperability, data quality, privacy, and security. The paper discusses state, action, reward, and constraints through a generic RL/MDP framing, but its strongest evidence concerns objective variation and practical validation barriers.

S2.10 Applications of reinforcement learning for building energy efficiency control: A review [10]

Table S2.47: RQ2 recoded audit table for Fu et al. (2022). Scores use the shared 0–3 evidence-intensity scale.

Survey	State	Action	Reward	Objectives	Constraints	Comparability problems	Recoded evidence justification
Fu et al. (2022)	2	2	2	2	2	3	The survey explicitly defines the MDP components used in RL for intelligent buildings, including state, action, reward, transition, and discount factor. State, action, and reward therefore receive score 2: they are explained and illustrated, but not systematically classified across reviewed studies as standalone taxonomies. Objectives receive score 2 because energy consumption, cost, comfort, and comfort-violation penalties are discussed through examples and case studies. Constraints receive score 2 in the formulation matrix because comfort penalties, safety, security, training cost, deployment cost, and real-building limitations are discussed, but not structurally taxonomized. Comparability problems receive score 3 because the paper strongly emphasizes practical limitations, especially simulation-to-real differences, long training time, real-world training cost, unsafe exploration, equipment risk, security concerns, and the need for more realistic deployment pathways.

Table S2.48: RQ2 formulation-consistency scores for Fu et al. (2022), recoded using the shared evidence-intensity scale.

Survey	State	Action	Reward	Objectives	Constraints
Fu et al. (2022)	2	2	2	2	2

Table S2.49: RQ2 comparability-barrier scores for Fu et al. (2022), recoded using the shared evidence-intensity scale.

Survey	State	Action	Reward	Objective/metric	Constraint	Baseline	Simulator/testbed
Fu et al. (2022)	1	2	2	2	3	0	3

Table S2.50: RQ2 summary table for Fu et al. (2022), after recoding.

Survey	Formulation strength	Biggest inconsistency identified	Main comparability consequence
Fu et al. (2022)	Moderate	Building RL studies share a generic MDP framing, but differ in how concrete control problems instantiate state, action, and reward and in how action-space structure is matched to algorithm families.	Cross-study comparison is weakened because many results depend on simulation setup, problem-specific state/action spaces, reward trade-offs, and deployment assumptions, while practical transfer is limited by long training times, real-world cost, unsafe exploration, equipment risk, and safety/security concerns.

Table S2.51: One-line RQ2 answer for Fu et al. (2022), after recoding.

Survey	RQ2 answer
Fu et al. (2022)	Fu et al. (2022) contribute to RQ2 mainly by giving a clear generic MDP framing and illustrative state–action–reward examples for building control. Its stronger contribution is to comparability barriers: the paper shows that simulation dependence, action-space and algorithm-suitability differences, training burden, unsafe real-world exploration, deployment cost, and safety/security concerns limit meaningful comparison and transfer across studies.

S2.11 Building energy management with reinforcement learning and model predictive control: A survey [11]

Table S2.52: RQ2 recoded audit table for Zhang et al. (2022). Scores use the shared 0–3 evidence-intensity scale.

Survey	State	Action	Reward	Objectives	Constraints	Comparability problems	Recoded evidence justification
Zhang et al. (2022)	2	2	2	2	2	3	The survey provides a technical background for BEMS, RL, and MPC, including MDP framing, observations, actions, rewards, objectives, and constraints. State receives score 2 because raw observations, state spaces, historical building data, and model-based RL are discussed, but state definitions are not systematically classified. Action receives score 2 because appliance scheduling, EV charging, HVAC setpoints, multi-agent RL, and continuous/discrete action settings are discussed. Reward receives score 2 because energy cost, comfort, demand response, and dissatisfaction are discussed through RL formulation and application examples. Objectives receive score 2 because energy consumption, comfort, cost, grid efficiency, stability, and demand response are discussed, but not as a standalone objective taxonomy. Constraints receive score 2 in the formulation matrix because MPC safety constraints, RL safety, data efficiency, time/memory complexity, robust adaptability, and online optimization are discussed. Comparability problems receive score 3 because the paper strongly emphasizes method-selection trade-offs between RL and MPC, simulator/model dependence, real-world exploration cost, data availability, safety handling, computational complexity, and generalization limitations.

Table S2.53: RQ2 formulation-consistency scores for Zhang et al. (2022), recoded using the shared evidence-intensity scale.

Survey	State	Action	Reward	Objectives	Constraints
Zhang et al. (2022)	2	2	2	2	2

Table S2.54: RQ2 comparability-barrier scores for Zhang et al. (2022), recoded using the shared evidence-intensity scale.

Survey	State	Action	Reward	Objective/metric	Constraint	Baseline	Simulator/testbed
Zhang et al. (2022)	2	2	2	2	3	2	3

Table S2.55: RQ2 summary table for Zhang et al. (2022), after recoding.

Survey	Formulation strength	Biggest inconsistency identified	Main comparability consequence
Zhang et al. (2022)	Moderate	Building-control studies share a generic formulation language, but differ in modeling assumptions, controller structure, safety treatment, data regime, simulator dependence, and whether the method is evaluated as RL, MPC, or a hybrid strategy.	Cross-study comparison is weakened because controller performance depends heavily on model fidelity, historical-data availability, safety and constraint handling, computational complexity, and practical assumptions about simulation, online optimization, and real-world exploration.

Table S2.56: One-line RQ2 answer for Zhang et al. (2022), after recoding.

Survey	RQ2 answer
Zhang et al. (2022)	Zhang et al. (2022) contribute to RQ2 mainly by comparing RL, MPC, and hybrid BEMS control strategies under practical modeling, data, safety, and deployment assumptions. The paper provides a generic formulation of state, action, reward, objectives, and constraints, but its strongest RQ2 contribution is showing that simulator dependence, model fidelity, safety handling, data efficiency, baseline/controller choice, and method-selection trade-offs make meaningful cross-study comparison and transfer difficult.

S2.12 Reinforcement learning: theory and applications in HEMS [12]

Table S2.57: RQ2 recoded audit table for Al-Ani & Das (2022). Scores use the shared 0–3 evidence-intensity scale.

Survey	State	Action	Reward	Objectives	Constraints	Comparability problems	Recoded evidence justification
Al-Ani and Das (2022)	2	2	2	3	2	2	The survey provides a generic RL/MDP formulation for HEMS, explicitly defining state space, action space, reward, return, and the MDP tuple. State, action, and reward therefore receive score 2: they are explained as part of the formulation, but are not systematically classified across reviewed studies. Objectives receive score 3 because the paper synthesizes HEMS applications around high-level goals such as energy cost, occupant comfort, load balance, and performance evaluation, and proposes common performance metrics. Constraints receive score 2 because appliance and system limitations, physical constraints, and MPC-related constraint handling are discussed, but not developed as a standalone taxonomy. Comparability problems receive score 2 because the paper contributes to cross-study comparison through common performance metrics and discussion of simulation versus real-world deployment, but does not make comparability breakdown the central structure of the review.

Table S2.58: RQ2 formulation-consistency scores for Al-Ani & Das (2022), recoded using the shared evidence-intensity scale.

Survey	State	Action	Reward	Objectives	Constraints
Al-Ani & Das (2022)	2	2	2	3	2

Table S2.59: RQ2 comparability-barrier scores for Al-Ani & Das (2022), recoded using the shared evidence-intensity scale.

Survey	State	Action	Reward	Objective/metric	Constraint	Baseline	Simulator/testbed
Al-Ani & Das (2022)	1	1	2	3	2	0	2

Table S2.60: RQ2 summary table for Al-Ani & Das (2022), after recoding.

Survey	Formulation strength	Biggest inconsistency identified	Main comparability consequence
Al-Ani & Das (2022)	Moderate, metric-oriented	HEMS studies are organized more clearly around broad optimization goals and performance metrics than around a detailed shared taxonomy of state, action, reward, and constraints.	Cross-study comparison is partially supported by common high-level objectives and proposed performance metrics, but remains limited because the review does not deeply standardize how studies define RL formulation components or validation settings.

Table S2.61: One-line RQ2 answer for Al-Ani & Das (2022), after recoding.

Survey	RQ2 answer
Al-Ani and Das (2022)	Al-Ani & Das (2022) contribute to RQ2 mainly by giving a generic RL/MDP formulation for HEMS and by structurally emphasizing common objectives and performance metrics, especially energy cost, occupant comfort, and load balance. The paper supports comparability more through metric-level synthesis than through a detailed cross-study taxonomy of state, action, reward, and constraints.

S2.13 A review of reinforcement learning for autonomous building energy management [13]

Table S2.62: RQ2 recoded audit table for Mason & Grijalva (2019). Scores use the shared 0–3 evidence-intensity scale.

Survey	State	Action	Reward	Objectives	Constraints	Comparability problems	Recoded evidence justification
Mason and Grijalva (2019)	2	2	2	2	2	3	The survey provides a clear generic RL/MDP formulation for building energy management and illustrates how state, action, and reward appear in applications such as HVAC, water-heater control, and home energy management. State, action, and reward therefore receive score 2: they are discussed and illustrated, but not systematically classified across reviewed studies. Objectives receive score 2 because energy cost, thermal comfort, peak reduction, and energy savings are discussed, including multi-objective trade-offs, but objectives are not organized as a standalone taxonomy. Constraints receive score 2 because real-building deployment limits, simulator fidelity, exploration cost, and robustness under changing real-world conditions are discussed. Comparability problems receive score 3 because the paper strongly emphasizes simulation dependence, simulator design, limited real-world validation, and uncertainty about robustness under practical conditions as major limitations for interpreting and comparing reported performance.

Table S2.63: RQ2 formulation-consistency scores for Mason & Grijalva (2019), recoded using the shared evidence-intensity scale.

Survey	State	Action	Reward	Objectives	Constraints
Mason & Grijalva (2019)	2	2	2	2	2

Table S2.64: RQ2 comparability-barrier scores for Mason & Grijalva (2019), recoded using the shared evidence-intensity scale.

Survey	State	Action	Reward	Objective/metric	Constraint	Baseline	Simulator/testbed
Mason & Grijalva (2019)	2	1	2	2	2	0	3

Table S2.65: RQ2 summary table for Mason & Grijalva (2019), after recoding.

Survey	Formulation strength	Biggest inconsistency identified	Main comparability consequence
Mason & Grijalva (2019)	Moderate	RL-for-building studies share a generic MDP formulation, but differ in state selection, reward/objective trade-offs, simulator design, and robustness assumptions.	Cross-study comparison is weakened because most evidence comes from simulation, and reported performance depends heavily on simulator fidelity, assumed disturbances, objective trade-offs, and whether the learned controller is robust under changing real-world conditions.

Table S2.66: One-line RQ2 answer for Mason & Grijalva (2019), after recoding.

Survey	RQ2 answer
Mason and Grijalva (2019)	Mason & Grijalva (2019) contribute to RQ2 mainly by providing a generic MDP framing and examples of state, action, and reward in building energy management applications. The paper’s strongest comparability contribution is its emphasis on state-selection difficulty, conflicting objectives, simulator design, simulation-heavy validation, and uncertain robustness under changing real-world conditions.

S2.14 Multi objectives reinforcement learning for smart buildings: A systematic review of algorithms, applications and future perspectives [14]

Table S2.67: RQ2 recoded audit table for Huynh et al. (2025). Scores use the shared 0–3 evidence-intensity scale.

Survey	State	Action	Reward	Objectives	Constraints	Comparability problems	Recoded evidence justification
Huynh et al. (2025)	2	2	3	3	2	3	The survey explains traditional RL/MDP and MORL/MOMDP formulation, including state, action, reward, and policy interaction. State and action therefore receive score 2: they are discussed in the formulation background, but they are not central cross-study taxonomies. Reward receives score 3 because vector-valued rewards, scalarization, Pareto-based strategies, weighted-sum rewards, constraint-based rewards, and dynamic reward adjustment are structurally central to the review. Objectives receive score 3 because the review is organized around common MORL objectives, objective co-occurrence, objective conflicts, application trends, and performance evaluation. Constraints receive score 2 because operational bounds, comfort limits, computational complexity, data quality, safety/security, and real-time implementation limits are discussed, but constraints are not developed as a standalone taxonomy. Comparability problems receive score 3 because the paper structurally discusses objective heterogeneity, reward scalarization, benchmarking methods, virtual versus real-world evaluation environments, limited code/open-data availability, weak reproducibility, and the need for standardized evaluation frameworks.

Table S2.68: RQ2 formulation-consistency scores for Huynh et al. (2025), recoded using the shared evidence-intensity scale.

Survey	State	Action	Reward	Objectives	Constraints
Huynh et al. (2025)	2	2	3	3	2

Table S2.69: RQ2 comparability-barrier scores for Huynh et al. (2025), recoded using the shared evidence-intensity scale.

Survey	State	Action	Reward	Objective/metric	Constraint	Baseline	Simulator/testbed
Huynh et al. (2025)	0	0	3	3	2	3	3

Table S2.70: RQ2 summary table for Huynh et al. (2025), after recoding.

Survey	Formulation strength	Biggest inconsistency identified	Main comparability consequence
Huynh et al. (2025)	Strong but specialized	MORL studies differ substantially in which objectives they optimize, how objectives interact or conflict, and how trade-offs are encoded through vector rewards, scalarization, reward design, and policy strategy.	Cross-study comparison is weakened because studies optimize different objective combinations, use different reward/scalarization strategies, compare against different baselines, and are evaluated under heterogeneous virtual or field environments with limited reproducibility, code sharing, and standardized metrics.

Table S2.71: One-line RQ2 answer for Huynh et al. (2025), after recoding.

Survey	RQ2 answer
Huynh et al. (2025)	Huynh et al. (2025) contribute to RQ2 mainly by structurally organizing MORL studies around objective heterogeneity, objective conflicts, vector-valued rewards, scalarization strategies, reward-design choices, benchmarking methods, and evaluation environments. The paper is especially useful for showing that differences in objective trade-offs, reward scalarization, baselines, reproducibility, and virtual versus real-world validation hinder direct cross-study comparability.

S2.15 An overview of reinforcement learning-based approaches for smart home energy management systems with energy storages [15]

Table S2.72: RQ2 recoded audit table for Pinthurat et al. (2024). Scores use the shared 0–3 evidence-intensity scale.

Survey	State	Action	Reward	Objectives	Constraints	Comparability problems	Recoded evidence justification
Pinthurat et al. (2024)	2	2	3	3	3	3	The survey provides a generic RL formulation for smart home energy management systems with energy storage, so state and action receive score 2: they are explained through observations, storage-related variables, controllable appliances, storage operation, electric vehicles, and grid interaction, but they do not structure the paper as standalone state/action taxonomies. Reward receives score 3 because reward function is an explicit comparative review criterion. Objectives receive score 3 because the paper structurally compares optimization goals such as cost, efficiency, renewable-energy integration, storage operation, grid stability, resilience, and energy optimization. Constraints receive score 3 because storage-related constraints are central comparative dimensions, including state of charge, state of health, life cycle, degradation, energy-storage constraints, cost, efficiency, operation mode, pricing mechanism, and renewable-energy integration. Comparability problems receive score 3 because the paper strongly emphasizes reward/objective/constraint heterogeneity, privacy and security concerns, scalability limits, hyperparameter sensitivity, sample inefficiency, and the need for real-world testing and validation to bridge theory and practice.

Table S2.73: RQ2 formulation-consistency scores for Pinthurat et al. (2024), recoded using the shared evidence-intensity scale.

Survey	State	Action	Reward	Objectives	Constraints
Pinthurat et al. (2024)	2	2	3	3	3

Table S2.74: RQ2 comparability-barrier scores for Pinthurat et al. (2024), recoded using the shared evidence-intensity scale.

Survey	State	Action	Reward	Objective/metric	Constraint	Baseline	Simulator/testbed
Pinthurat et al. (2024)	1	1	3	3	3	0	3

Table S2.75: RQ2 summary table for Pinthurat et al. (2024), after recoding.			
Survey	Formulation strength	Biggest inconsistency identified	Main comparability consequence
Pinthurat et al. (2024)	Strong but specialized	Smart-home energy-management studies with energy storage differ substantially in reward design, optimization goals, storage assumptions, and whether storage properties and constraints such as SoC, SoH, degradation, life cycle, cost, efficiency, operation mode, and pricing mechanism are explicitly modeled.	Cross-study comparison is weakened because studies are evaluated under different storage assumptions, pricing mechanisms, operation modes, reward/objective formulations, and validation settings, while practical deployment remains limited by privacy/security concerns, scalability issues, and the lack of real-world testing.

Table S2.76: One-line RQ2 answer for Pinthurat et al. (2024), after recoding.	
Survey	RQ2 answer
Pinthurat et al. (2024)	Pinthurat et al. (2024) contribute to RQ2 mainly by structurally comparing reward functions, objectives, and storage-related constraints across smart-home energy-management studies with energy storage. The paper is especially useful for showing that storage modeling choices, reward design, operation mode, pricing mechanism, privacy/security concerns, scalability, and limited real-world validation hinder direct cross-study comparability.

S2.16 Reinforcement learning for control and optimization of real buildings: Identifying and addressing implementation hurdles [16]

Table S2.77: RQ2 recoded audit table for Kannari et al. (2025). Scores use the shared 0–3 evidence-intensity scale.							
Survey	State	Action	Reward	Objectives	Constraints	Comparability problems	Recoded evidence justification
Kannari et al. (2025)	2	3	3	3	2	3	The survey contributes to RQ2 mainly through real-building deployment evidence. State receives score 2 because the paper gives a generic RL framing in which policies map perceived states to actions and because weather, occupancy, pretraining conditions, and building context are discussed, but state is not the main standalone taxonomy. Action receives score 3 because real-building action interfaces are structurally compared through control interval, output resolution, actuator step size, direct control signals, supervisory setpoint adjustment, planning-style actions, and real equipment constraints. Reward receives score 3 because reward and comfort formulations are structurally compared across real-building studies, including temperature penalties, PPD-based comfort, personalized comfort, air-movement acceptability, PM2.5-aware formulations, and penalties for frequent action changes. Objectives receive score 3 because optimization target and comfort metric are visible comparison dimensions across deployed studies. Constraints receive score 2 because actuator limits, control intervals, comfort ranges, safety checks, fallback control, pretraining, damage risk, occupant dissatisfaction, human oversight, privacy, security, and reliability are discussed, but not developed as a standalone taxonomy. Comparability problems receive score 3 because the paper explicitly argues that real-building controller comparison is not straightforward due to differences in weather, occupancy, pretraining setup, safety procedures, deployment mode, control interval, field-test conditions, and validation maturity.

Table S2.78: RQ2 formulation-consistency scores for Kannari et al. (2025), recoded using the shared evidence-intensity scale.

Survey	State	Action	Reward	Objectives	Constraints
Kannari et al. (2025)	2	3	3	3	2

Table S2.79: RQ2 comparability-barrier scores for Kannari et al. (2025), recoded using the shared evidence-intensity scale.

Survey	State	Action	Reward	Objective/metric	Constraint	Baseline	Simulator/testbed
Kannari et al. (2025)	2	3	3	3	2	0	3

Table S2.80: RQ2 summary table for Kannari et al. (2025), after recoding.

Survey	Formulation strength	Biggest inconsistency identified	Main comparability consequence
Kannari et al. (2025)	Strong but specialized	Real-building RL studies differ substantially in action interface, reward and comfort formulation, optimization target, deployment mode, control interval, safety procedure, and validation duration.	Cross-study comparison is weakened because field studies are conducted under different weather and occupancy conditions, different pretraining and safety arrangements, different control intervals and deployment modes, and different comfort/reward formulations, so performance cannot be compared directly without contextual normalization.

Table S2.81: One-line RQ2 answer for Kannari et al. (2025), after recoding.

Survey	RQ2 answer
Kannari et al. (2025)	Kannari et al. (2025) contribute to RQ2 mainly by showing that real-building RL studies vary in action interface, reward and comfort design, optimization target, deployment mode, control interval, safety procedures, and validation practice. These differences make safe deployment and meaningful cross-study comparison difficult even when studies share a generic RL formulation language.

S2.17 Dynamic indoor thermal environment using reinforcement learning-based controls: Opportunities and challenges [17]

Table S2.82: RQ2 recoded audit table for Chatterjee & Khovalyg (2023). Scores use the shared 0–3 evidence-intensity scale.

Survey	State	Action	Reward	Objectives	Constraints	Comparability problems	Recoded evidence justification
Chatterjee and Khovalyg (2023)	2	3	2	3	2	3	The survey contributes to RQ2 in a specialized way by connecting RL-based HVAC control to dynamic indoor thermal environments. State receives score 2 because indoor-environment variables, occupancy, weather, building context, and controlled parameters are discussed, but state is not developed as a standalone taxonomy. Action receives score 3 because action-space design is structurally discussed through HVAC control parameters, continuous versus discrete action spaces, temperature setpoints, HVAC on/off decisions, and controlled variables. Reward receives score 2 because reward-function quality and energy/comfort outcomes are discussed as part of RL-control performance, but reward is not the main taxonomy. Objectives receive score 3 because the paper structurally emphasizes energy-use reduction, thermal comfort, health, and dynamic indoor-environment control as central goals. Constraints receive score 2 because acceptable temperature ranges, step-change limits, temperature-drift limits, ASHRAE/ISO comfort restrictions, security, training burden, and real-world feasibility are discussed, but not as a standalone taxonomy. Comparability problems receive score 3 because the paper strongly emphasizes heterogeneous co-simulation environments, variation in controlled parameters and HVAC systems, limited real-building implementation, lack of a shared virtual testbed, and the need for open interoperable benchmarking infrastructure.

Table S2.83: RQ2 formulation-consistency scores for Chatterjee & Khovalyg (2023), recoded using the shared evidence-intensity scale.

Survey	State	Action	Reward	Objectives	Constraints
Chatterjee & Khovalyg (2023)	2	3	2	3	2

Table S2.84: RQ2 comparability-barrier scores for Chatterjee & Khovalyg (2023), recoded using the shared evidence-intensity scale.

Survey	State	Action	Reward	Objective/metric	Constraint	Baseline	Simulator/testbed
Chatterjee & Khovalyg (2023)	1	3	2	3	2	1	3

Table S2.85: RQ2 summary table for Chatterjee & Khovalyg (2023), after recoding.

Survey	Formulation strength	Biggest inconsistency identified	Main comparability consequence
Chatterjee & Khovalyg (2023)	Strong but specialized	RL-for-dynamic-indoor-environment studies differ substantially in action-space design, controlled variables, HVAC systems, co-simulation environments, and the balance among energy, comfort, and health objectives.	Cross-study comparison is weakened because studies are evaluated under different simulation platforms, HVAC configurations, controlled parameters, action spaces, and validation settings, while only a minority are implemented in real buildings and the field lacks a shared open interoperable virtual testbed for benchmarking.

Table S2.86: One-line RQ2 answer for Chatterjee & Khovalyg (2023), after recoding.

Survey	RQ2 answer
Chatterjee and Khovalyg (2023)	Chatterjee & Khovalyg (2023) contribute to RQ2 mainly by showing that RL studies for dynamic indoor thermal environments vary in action-space design, controlled parameters, HVAC systems, objectives, and simulation environments. The paper is especially useful for showing that limited real-building deployment and the lack of an open interoperable virtual testbed make meaningful cross-study comparison difficult.

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